



Executive Overview: Portfolio Risk



GENDER

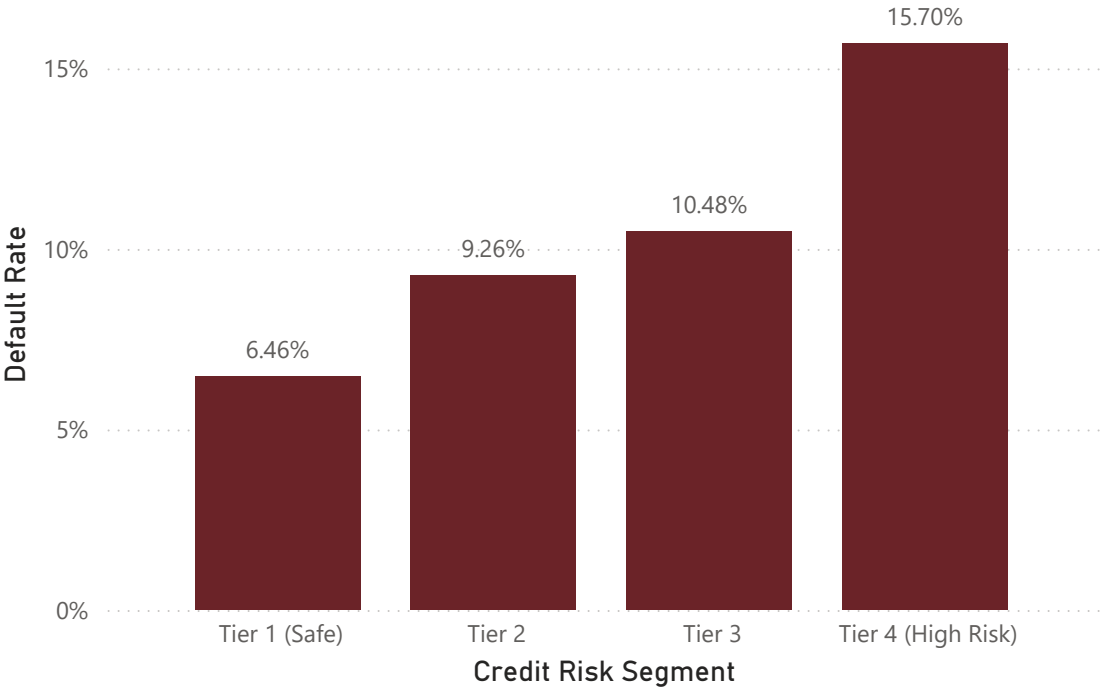
All

308K
Total Apps

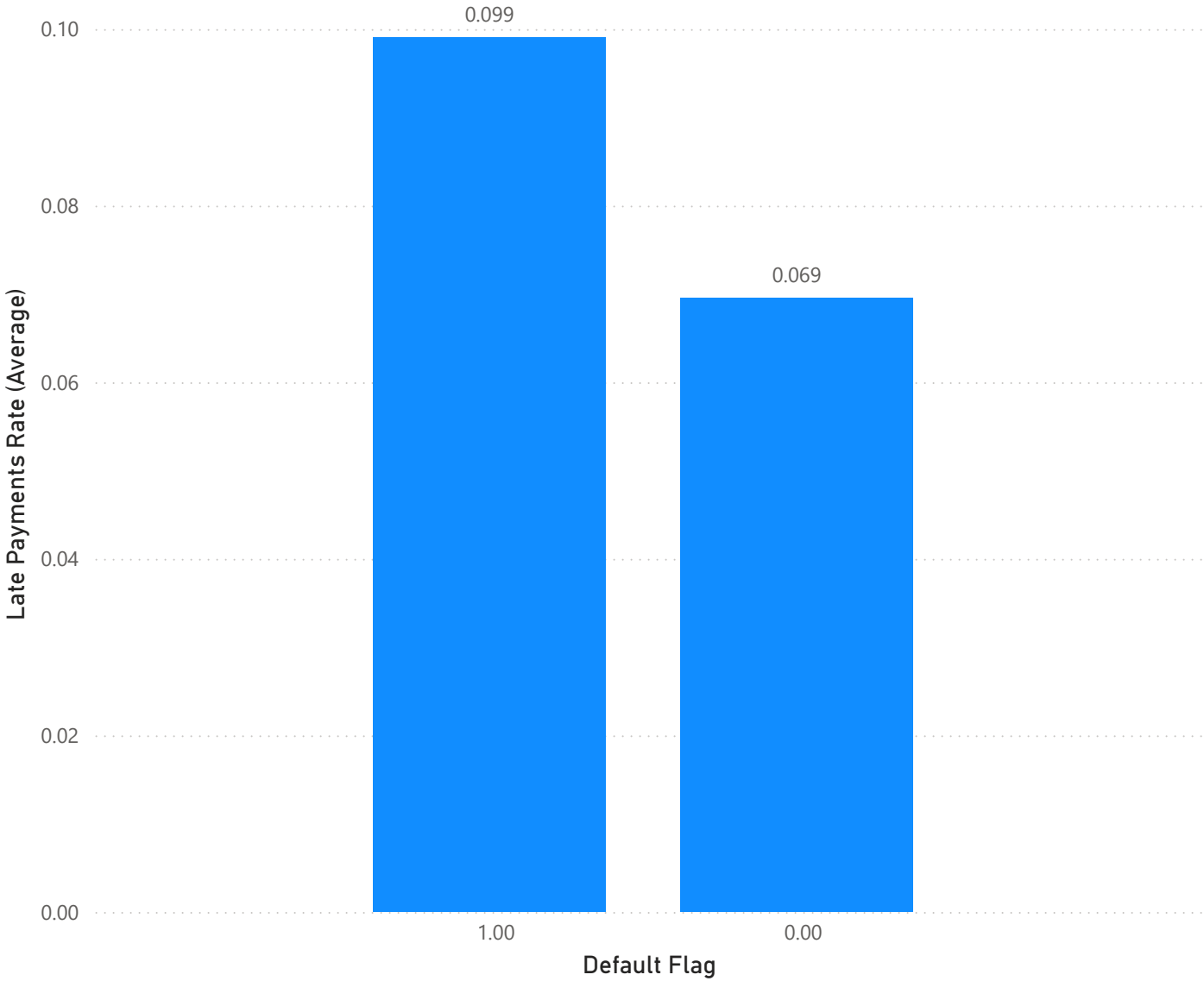
8.07%
Default Rate

\$599.03K
Avg Loan Amount

Default Rate by Risk Tier



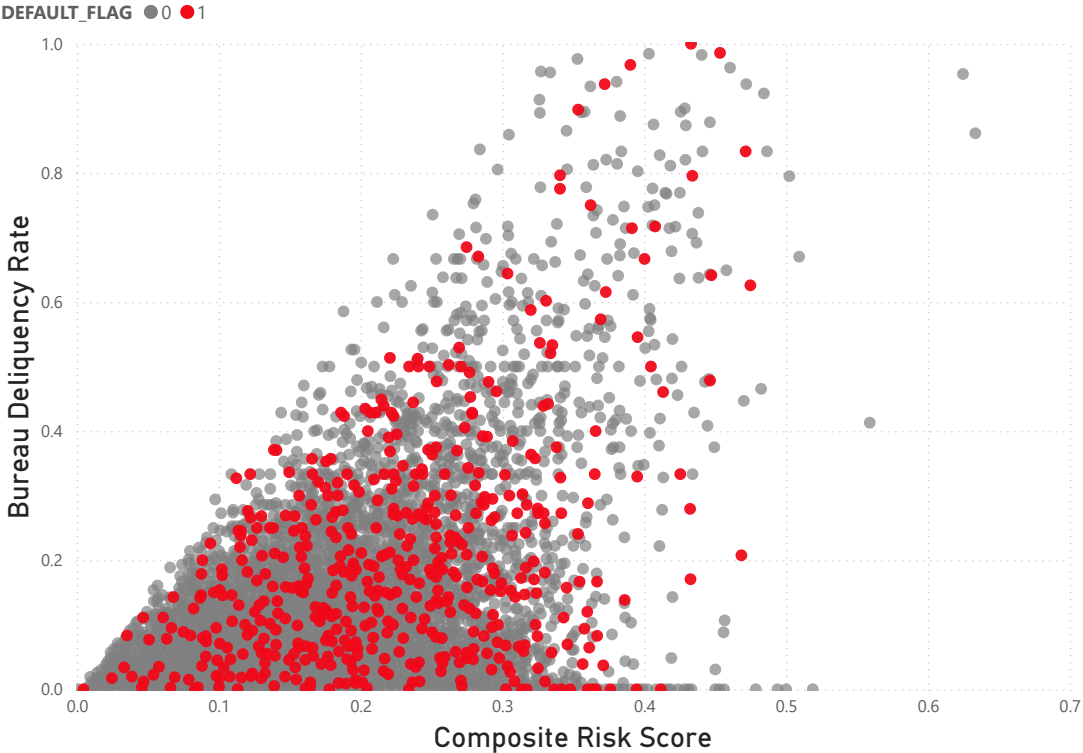
Average Installment Late Rate: Repaid vs. Defaulted



Historical payment behavior is one of the strongest predictors of future defaults. Our analysis reveals that applicants who ultimately default miss nearly 10% of their past installment payments, compared to a 7% baseline for safe borrowers.

While a 3% absolute difference may seem small, it represents a nearly 45% relative increase in missed payments. The XGBoost model heavily relies on this behavioral gap to identify hidden risk in applicants who might otherwise have acceptable external credit scores.

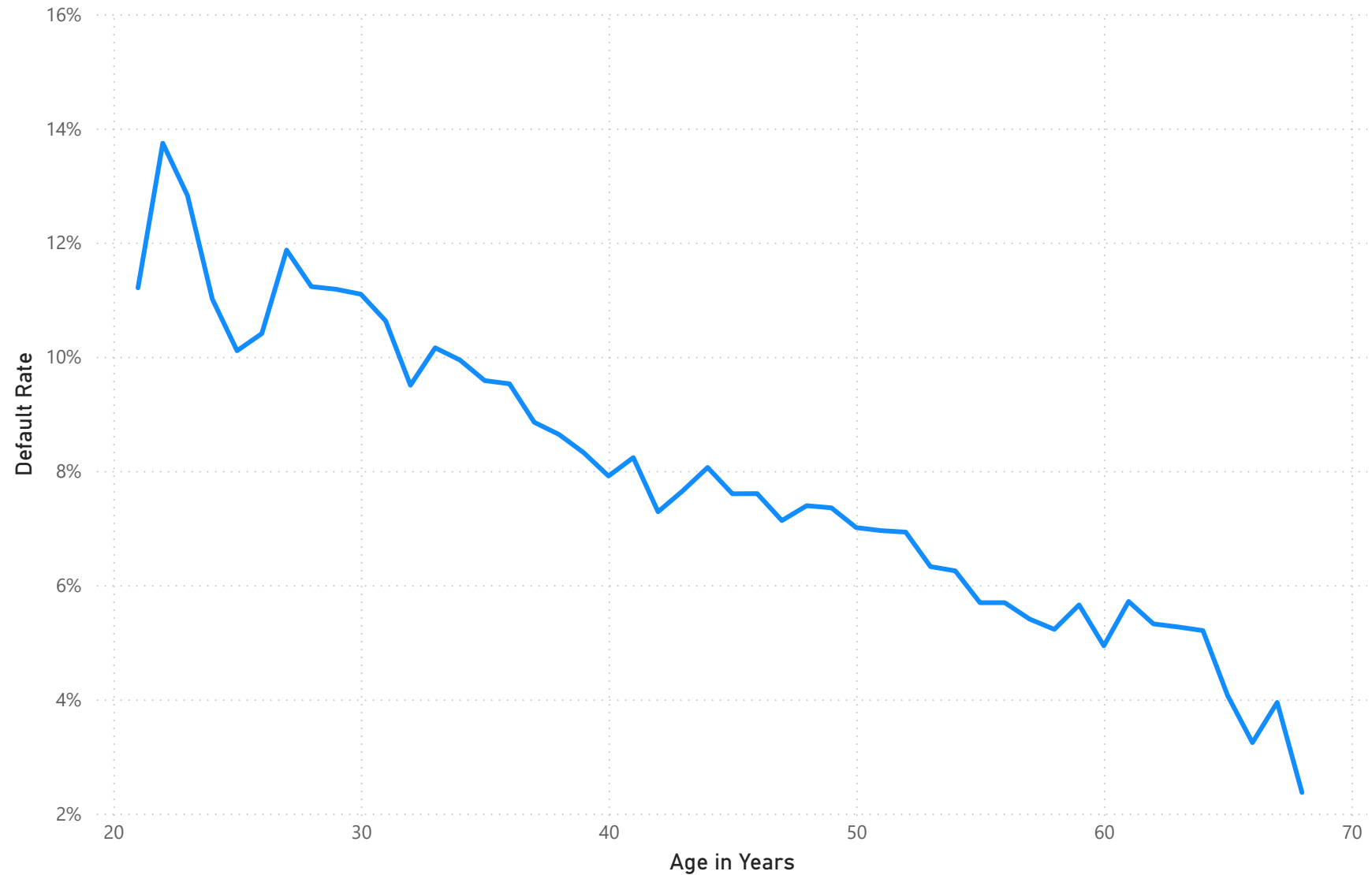
Default Clusters: Composite Risk Score vs. Bureau Delinquency



This scatter plot maps our Composite Risk Score against Bureau Delinquency. While we see a clear, dangerous cluster of defaulters (red) in the upper-right quadrant, there is also a notable density of unexpected defaulters in the "safe" bottom-left zone.

This visual highlights two things: First, elevated risk scores and delinquency histories are near-guaranteed predictors of default. Second, simple heuristic rules are not enough to catch everyone. Sudden life events cause "safe-looking" applicants to default, which is exactly why we deployed an XGBoost Machine Learning model—to catch the complex, multi-variate hidden risks that a simple 2D scatter plot misses.

Default Rate Vs Age Graph



Borrowers in their early 20s exhibit the highest financial volatility, peaking at a default rate near 14%. Risk steadily declines as applicants age, crossing below the 8% baseline in their late 30s and dropping to just 2.3% by retirement age.